

Airspace Access and Integration

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Low Altitude Economics

Week 8

Learning Objectives

By the end of this class, you should be able to:

- **Explain** how safe operating capacity and demand create airspace congestion.
- **Compare** queues, reservations, priority rules, peak pricing, and auctions as allocation mechanisms.
- **Estimate** a simple delay externality and an operator's maximum affordable slot payment.
- **Evaluate** how digital coordination and interoperability affect cost, safety, and scalability.
- **Design** a phased airspace-integration roadmap with explicit evidence and decision gates.

The 8:00 AM Capacity Conflict

Imagine that two operations request the same constrained corridor and time window.

- One is a time-sensitive public-service delivery.
- The other is a scheduled commercial passenger or cargo operation.

Only one request can be served safely within the available window.

- Should priority depend on arrival time, reservation, social value, price, or a predefined public-service rule?
- **Quick poll (2 min):** Which allocation rule would you choose, and what problem might it create?

Session 1

Congestion Pricing, Slot Auctions & Mechanism Design

Concept Primer: The Airspace Bottleneck

Airspace capacity is not simply a fixed number of drones per minute.

- Safe capacity depends on route geometry, separation, aircraft performance, surveillance, communications, weather, contingency routes, human oversight, and interactions with other airspace users.
- Vertiport capacity, charging, passenger processing, and available stands may become binding before the corridor itself.
- Demand above reliable capacity creates delay, displacement, cancellation, or the need for another route or time.
- Sandbox X includes controlled trials related to airspace management and simultaneous unmanned-aircraft operations, but does not establish a general published capacity for Victoria Harbour corridors.

Managerial lesson: Define the constrained resource and time interval before designing an allocation mechanism.

Concept Primer: Congestion Externalities

A congestion externality arises when one additional operation increases delay, risk-management effort, or disruption for other users without bearing the full resulting cost.

- A congestion charge aims to reflect the additional cost imposed on others.
- The relevant cost may include aircraft operating cost, passenger or cargo delay, schedule disruption, and additional coordination effort.
- Pricing is only one response. Reservations, quotas, priority rules, rerouting, and capacity investment may also be appropriate.
- Some public-service missions should be governed by explicit priority rules rather than willingness to pay.

Applied Tool: A Simple Marginal Congestion Cost

Illustrative Calculation

$$\text{Marginal Congestion Cost} = \sum_j (\text{Additional Delay}_j \times \text{Value of Delay}_j) \\ + \text{Other Incremental Costs}$$

- Sum the incremental delay imposed on all affected users, not only the next aircraft.
- Values may differ across passengers, cargo, aircraft, public services, and schedule commitments.
- The theoretically efficient charge equals marginal external cost, but that cost is difficult to observe precisely in real time.
- A practical peak charge may therefore use estimated bands and periodic review.

Back-of-the-Envelope: Estimating a Delay Externality

Scenario: One additional operation delays three other flights by five minutes each.

Affected Users

- Flight 1 delay value: HK\$100/min
- Flight 2 delay value: HK\$60/min
- Flight 3 delay value: HK\$40/min
- Total value: HK\$200/min

Estimated External Cost

- Delay: 5 minutes
- External delay cost: 5×200
- **HK\$1,000**
- A peak charge near this amount may encourage a flexible operation to move, but its response is not guaranteed.

Takeaway: A congestion charge should reflect the total incremental cost imposed on others, while priority public-service operations require separate rules.

Illustrative assumptions; excludes safety, environmental, and schedule-recovery costs.

Session 2

Slot Auctions, Mechanisms & Network Externalities

Concept Primer: Mechanism Design and Slot Allocation

Mechanism design asks how rules shape participants' information, incentives, and outcomes.

- **Administrative rules:** schedules, rotation, first-come-first-served, historical rights, or public-service priority.
- **Posted prices:** predictable peak and off-peak charges.
- **Auctions:** participants submit bids for scarce time windows or route access.
- **Package allocation:** linked departure, corridor, arrival, and charging reservations may need to be allocated together.

Design criteria: safety, efficiency, fairness, competition, transparency, reliability, simplicity, and access for new entrants. *Failure to obtain a linked slot should lead to rejection or rescheduling, not an unsafe flight.*

Applied Tool: Maximum Affordable Slot Payment

A Simple Operator Screen

Maximum Affordable Payment = Incremental Revenue—Incremental Operating Costs

- Use incremental values associated with obtaining the slot, not total company revenue and OPEX.
- The required contribution may cover fixed cost, risk, or a target return.
- The rational bid also depends on auction rules, expected rival bids, alternative slots, and the risk that linked resources are unavailable.
- A high bid does not automatically imply high social value or justify exclusion of smaller operators.

Back-of-the-Envelope: Comparing Affordable Bids

Scenario: Two operators value the same commercial slot.

Operator A

- Incremental revenue: HK\$2,000
- Incremental cost: HK\$1,200
- Required contribution: HK\$300
- **Maximum affordable payment: HK\$500**

Operator B

- Incremental revenue: HK\$2,000
- Incremental cost: HK\$600
- Required contribution: HK\$300
- **Maximum affordable payment: HK\$1,100**

Takeaway: Operator B can afford a higher bid under these assumptions. The actual payment depends on the auction design and competing bids, not necessarily HK\$501.

Illustrative assumptions; commercial slots remain subject to safety, priority, and competition rules.

Concept Primer: Network Effects and Shared Information

A network effect exists when participation by additional compatible users increases the value of the network to existing users.

- Common interfaces and timely information may improve route planning, conflict detection, situational awareness, and coordination.
- More participants can also increase congestion, cybersecurity exposure, monitoring needs, and system complexity.
- Shared data does not automatically make a system safer. Data quality, latency, coverage, governance, and operational use matter.
- Some benefits are network effects; others are economies of scale or standardization benefits. They should not be treated as identical.

Applied Tool: Coordination-Cost Savings

A Simple Comparison

$$\text{Daily Net Saving} = \text{Cost of the Less-Integrated Process} \\ - \text{Cost of the Interoperable Process}$$

- Each process should include fixed systems, usage fees, communications, human oversight, cybersecurity, integration, certification, and exception handling.
- Interoperability can reduce duplicate interfaces and repeated manual data entry.
- It may also create migration cost, common dependencies, governance disputes, and switching risks.
- Avoid assuming that API or digital-services cost is near zero.

Back-of-the-Envelope: Two Coordination Processes

Assumptions: 1,000 operations/day and 30 operating days/month.

Less-Integrated Process

- Manual and system cost:
HK\$30/operation
- Daily cost: HK\$30,000
- **Monthly cost: HK\$900,000**

Interoperable Digital Process

- Fixed system and oversight:
HK\$12,000/day
- Variable cost: HK\$3/operation
- Daily cost: HK\$15,000
- **Monthly cost: HK\$450,000**

Takeaway: The interoperable process saves HK\$450,000 per month under these assumptions, but only if it achieves the required reliability, security, and service quality.

Illustrative assumptions; not a Hong Kong UTM or European U-space cost estimate.

Session 3

Interoperability, Roadmaps & GBA Integration

Concept Primer: Interoperability and Governance

Interoperability is the ability of authorized systems and organizations to exchange, interpret, and use information consistently.

- It may rely on common data models, interfaces, identity and access controls, service levels, and conformance testing.
- Shared standards can have public-good characteristics, but the systems, operations, cybersecurity, and assurance remain costly.
- Open standards do not require unrestricted public access to operational data.
- Governance must address data minimization, confidentiality, liability, competition, resilience, vendor dependence, and incident response.

Managerial lesson: Technical compatibility is necessary but not sufficient; institutions and incentives must also align.

Concept Primer: Airspace-Integration Roadmap

Integration should proceed through evidence-based stages rather than fixed promises of full scale by a particular year.

- **Stage 1: Defined trials** with limited routes, aircraft, times, users, and contingency procedures.
- **Stage 2: Extended operations** with more conditions, repeated performance, interoperable services, and stakeholder feedback.
- **Stage 3: Scaled network** only after safety, reliability, capacity, governance, and commercial thresholds are met.
- Cross-boundary development also requires coordination on customs, immigration, quarantine, spectrum, data, security, liability, and emergency response.

Applied Tool: Phased Roadmap Value

A Simple NPV Framework

$$\text{Roadmap NPV} = \sum_{t=0}^T \frac{\text{Incremental Cash Inflow}_t - \text{Incremental Cash Outflow}_t}{(1+r)^t}$$

- Include implementation cost in the year in which it occurs, rather than subtracting all costs outside the discounting formula.
- Cash flows should reflect probability of reaching each phase, operating limits, demand, cost, and possible abandonment.
- Decision gates preserve flexibility to pause, redesign, or expand when evidence changes.
- A positive NPV under assumptions does not prove that the wider market is viable.

Back-of-the-Envelope: A Three-Year Phased Roadmap

Assumptions: Discount rate = 10%. Net cash flows already combine phase revenue and implementation or operating cost.

Cash Flows

- Year 0 trial investment: –HK\$50M
- Year 1 net cash flow: HK\$10M
- Year 2 net cash flow: HK\$40M
- Year 3 net cash flow: HK\$50M

Present Values

- Year 1: HK\$9.1M
- Year 2: HK\$33.1M
- Year 3: HK\$37.6M
- Total PV of future net cash flows: HK\$79.8M
- **Roadmap NPV: HK\$29.8M**

Takeaway: The roadmap is attractive under these assumed cash flows. Probabilities, decision gates, terminal value, and cross-boundary dependencies still require analysis.

Illustrative assumptions; not a forecast of Hong Kong-GBA operations.

In-Class Exercise: Allocating a Scarce Slot

Scenario: Disruption reduces three planned corridors to one available commercial slot.

- **Roles:** Commercial operators, public-service customer, regulator or airspace manager, and community representative
- **Task:**
 1. calculate each commercial operator's maximum affordable payment;
 2. choose an allocation mechanism and explain its objective;
 3. apply a predefined public-service priority rule if triggered;
 4. state how cancelled or displaced operations are treated; and
 5. assess effects on efficiency, fairness, competition, and daily breakeven.
- **Timing:** 15 minutes plus 5-minute discussion.

Exercise Template: Allocation Ledger

Request	Incremental revenue	Incremental cost	Required contribution
Cargo operator	HK\$800	HK\$400	HK\$100
Passenger operator	HK\$2,000	HK\$600	HK\$400
Priority public service	Apply predefined eligibility and dispatch rule		

- Cargo maximum affordable payment: HK\$300.
- Passenger maximum affordable payment: HK\$1,000.
- If priority is triggered, record who bears the displacement cost and whether compensation, rescheduling, or reserved capacity applies.

Key Takeaways

- Airspace capacity depends on the complete operating system and cannot be reduced to an unsupported fixed number of drones per minute.
- Congestion pricing estimates external delay cost, while queues, reservations, priority rules, and auctions serve different objectives.
- Auctions can reveal willingness to pay but require safeguards for safety, competition, linked resources, new entrants, and public-service access.
- Interoperability may reduce duplicate coordination and support scale, but does not make digital services free or automatically safe.
- Network participation can create benefits and costs; data quality, cybersecurity, governance, and congestion remain essential.
- A credible roadmap links each expansion in capacity or operating privilege to evidence, review, and a financially consistent decision gate.

Week 8 Assignment Briefing

Airspace Allocation and Congestion Memo (maximum 1,200 words)

- **Due:** Beginning of Week 9 (group assignment)
- **Task:** Design an allocation mechanism for a constrained Hong Kong corridor or facility.
- **Requirements:**
 1. define the constrained resource, safe capacity, time interval, and demand scenario;
 2. compare at least two mechanisms, such as queueing, reservations, peak pricing, or auction;
 3. estimate a delay externality or maximum affordable payment using transparent assumptions;
 4. specify eligibility and treatment for priority public-service operations;
 5. model one weather or capacity shock and identify who bears displacement cost;
 6. explain the required digital information, interoperability, cybersecurity, and governance arrangements; and
 7. propose a phased implementation and evaluation plan.
- **Grading focus:** Economic reasoning (35%), operational realism (25%), welfare and competition analysis (20%), governance and

Economic and Social Impact of LAE

Preview question for next week:

We've spent 8 weeks figuring out how companies make money... but does the Low Altitude Economy actually make life better for the everyday citizens of Hong Kong?

Next Week Preview:

The Big Picture: Jobs, GDP, and the Environment

Pre-Class Readings Reminder (for Week 9):

- *NASA UAM Market Study (Network design & airspace)*
- *EASA Easy Access Rules for U-space (Interoperability)*

Questions?

(Final 15 minutes reserved for extended Q&A)